A Sparse-Cardinality Curvature Bound under Persistent Adversarial Noise

ZIJIAN ZENG, UCSI University, Malaysia
JIAYIN CAO, Ningbo University, China
WEINAN DAI, Zhejiang University, China
RUI WANG, Nanchang Institute of Technology, China
FEI DING, Alibaba Group, China

We study cardinality-constrained maximization of a monotone submodular function when a fixed value oracle has adversarial multiplicative error. The objective is the true function value, and queries may have any size. For exact total curvature $0 \le \kappa < 1$ and noise $0 < \varepsilon < 1$, we calibrate a hidden-set construction to give an explicit lower bound against adaptive randomized query algorithms. If $k = \lfloor n^\alpha \rfloor$ for fixed $0 < \alpha < 1/2$ and $\varepsilon_n k \to \infty$, every algorithm family with a fixed polynomial query bound has fixed valid instances on which its expected approximation is at most $\frac{1-\varepsilon_n}{1+\varepsilon_n}(1-\kappa) + o(1)$. The standard method of choosing the largest noisy singletons guarantees $\frac{1-\varepsilon_n}{1+\varepsilon_n}(1-\kappa+\kappa/k)$, giving additive asymptotic matching in this regime. For vanishing noise with $\varepsilon_n k \to \infty$, the limiting factor is $1-\kappa$, separated from the exact-oracle benchmark $1-\kappa/e$ when $\kappa > 0$.

Zijian Zeng 1002266693@ucsiuniversity.edu.my
Jiayin Cao 2795820153@qq.com
Weinan Dai 21725058@zju.edu.cn
Rui Wang 2002007@nut.edu.cn
Fei Ding dingfei@email.ncu.edu.cn

1 Model and contribution

Curvature measures how much an element's marginal value can fall as a set grows. With exact value access, this structure improves the approximation factor for monotone submodular maximization to $1 - \kappa/e$, up to arbitrary accuracy slack [SVW14, Theorems 6.1 and 7.1]. Persistent adversarial noise can remove much of that benefit. We identify a sparse-cardinality regime in which even unrestricted computation and adaptive randomized queries cannot asymptotically improve a simple singleton method.

Let $V = [n]$ and $n \geq k \geq 2$. The true function $f : 2^V \to \mathbb{R}_{\geq 0}$ is normalized, monotone and submodular, and $f(\{i\}) > 0$ for every i. Write $f(i \mid T) = f(T \cup \{i\}) - f(T)$ for $i \notin T$ and define its total curvature by

$$\kappa(f) = 1 - \min_{i \in V} \frac{f(i \mid V \setminus \{i\})}{f(\{i\})}.$$

A valid oracle is a single map $g : 2^V \to \mathbb{R}$, fixed before the algorithm's private random choices, such that

$$(1 - \varepsilon) f(T) \leq g(T) \leq (1 + \varepsilon) f(T) \qquad (T \subseteq V). \tag{1}$$

Repeated queries return the same value. Errors across different sets may be arbitrarily correlated; g need not be submodular or monotone. Queries of every cardinality are permitted. The output A must satisfy $|A| \leq k$, and approximation is measured by $f(A)/\mathrm{OPT}(f, k)$, where $\mathrm{OPT}(f, k) = \max_{|T| \leq k} f(T)$. Algorithms may know $n, k, \kappa, \varepsilon$ and use arbitrary computation. A polynomial query family has a bound Cn^d for constants C, d fixed for that family. We impose no bit-complexity or numerical precision bound.

Theorem 1 (Finite query bound). *Let n, k, r be integers with $n > k + r$ and $1 \leq r < k$. For $0 \leq \kappa < 1$ and $0 < \varepsilon < 1$, set*

$$\rho = \frac{1 - \varepsilon}{1 + \varepsilon}, \qquad a = \rho(1 - \kappa), \qquad b = \rho\kappa, \qquad c = \frac{(1 - \rho)(1 - \kappa)}{1 - a}. \tag{2}$$

Suppose $cn > ek$. For every possibly randomized adaptive algorithm making at most Q queries on every valid input and returning a feasible set, there is a fixed valid pair (f_H, g_H), independent of its private seed, with $\kappa(f_H) = \kappa$ and $\mathrm{OPT}(f_H, k) = k$, such that

$$\frac{\mathbb{E}[f_H(A)]}{k} \leq a + \frac{br}{k} + \frac{(1 - a)k}{n} + \min\left\{1, Q\left(\frac{ek}{cn}\right)^{cr}\right\}. \tag{3}$$

Here $H \subseteq V$ has size k, and the pair is explicitly

$$f_H(S) = |S \cap H| + a|S \setminus H| + b \min\{|S \setminus H|, r\}, \tag{4}$$

$$g_H(S) = \max\left\{(1 + \varepsilon)[a|S| + b \min\{|S|, r\}], (1 - \varepsilon) f_H(S)\right\}. \tag{5}$$

The expectation is only over the algorithm's randomness.

Almost-sure query and feasibility promises suffice. Since the proof uses only finitely many valid inputs at each n, their exceptional null sets of seeds can be discarded together.

Corollary 2 (Sparse cardinality). *Fix $0 < \alpha < 1/2$ and $0 \leq \kappa < 1$, let $k_n = \lfloor n^\alpha \rfloor$, and suppose $0 < \varepsilon_n < 1$ with $\varepsilon_n k_n \to \infty$. Put $\rho_n = (1 - \varepsilon_n)/(1 + \varepsilon_n)$. For every algorithm family using at most Cn^d queries, there is a sequence of fixed valid pairs of exact curvature κ for which*

$$\frac{\mathbb{E}[f_n(A_n)]}{\mathrm{OPT}(f_n, k_n)} \leq \rho_n(1 - \kappa) + o(1). \tag{6}$$

On every valid input, selecting the k_n largest noisy singleton values guarantees $\rho_n(1 - \kappa + \kappa/k_n)$. The bounds match additively; they match relatively when ρ_n is bounded away from zero.

The instances may depend on the algorithm family, but each oracle is fixed independently of its private seed. The statement quantifies separately over each fixed polynomial query bound. It does not optimize over unbounded polynomial degrees at a fixed n. For example, $\kappa = 1/2$ and $\varepsilon = 1/3$ yield the matching limiting factor $1/4$. If $\varepsilon_n \to 0$ while $\varepsilon_n k_n \to \infty$, the factor tends to $1 - \kappa$; for $\kappa > 0$ this is separated from the exact-oracle benchmark.

Relation to prior work. Horel and Singer [HS16, Section 2.1 and Proposition 8] study an approximately submodular objective F satisfying the same pointwise sandwich as (1). They supply both the hidden capped-rank pattern and the largest-noisy-singleton method. Their curvature factor $(1-\kappa)\rho^2$ is stated for the approximate objective F. Directly analyzing the same singleton method against true f gives one factor of ρ, as proved below. That normalization and its κ/k refinement are elementary baselines. Sviridenko, Vondrák and Ward [SVW14, Section 7] use a uniform modular shift to tune curvature in exact-oracle lower bounds. Hassidim and Singer [HS17, Theorem 8 and Section 5] already establish adversarial-noise hardness for randomized algorithms. Their displayed hardness regime does not prescribe the exact curvature below one used here. Persistence also appears in stochastic-noise algorithms: Bhawalkar, Cai, Feng, Liaw and Lin [BCF+25, Section 2] require unbiased noise and independence across distinct sets, assumptions absent from (1).

The calibration in (2) and finite bound (3) yield (6) for every sequence $\varepsilon_n k_n \to \infty$ in the stated regime. All-query-size leakage control is needed because queries are unrestricted. The construction and transcript strategy build on the preceding work.

2 Construction and leakage

2.1 Global oracle validity and exact curvature

Use the parameters of Theorem 1. For $s = |S|$, $h = |S \cap H|$ and $t = s - h$, define the baseline

$$F_0(S) = as + b\min\{s, r\}, \qquad g_0(S) = (1+\varepsilon)F_0(S). \tag{7}$$

All formulas define values on every subset before an algorithm is run. The marginal of an element of H in f_H is always one. Outside H, it is $a + b = \rho$ while fewer than r outside elements are present, and a thereafter. These marginals are positive and diminishing, so f_H is normalized, monotone and submodular. Since $n - k > r$, the full-set marginal of an outside element is a.

Table 1. Calibration of the hard family. The noisy intervals of the two singleton types meet at $1-\varepsilon$; both can receive that same answer.

Element type	True singleton	Full-set marginal	Marginal/singleton
Inside H	1	1	1
Outside H	ρ	$\rho(1-\kappa)$	$1-\kappa$

Table 1 proves that $\kappa(f_H) = \kappa$, including $\kappa = 0$. Also $f_H(S) \le h + (a+b)t \le s$ and $f_H(H) = k$, so the feasible optimum is exactly k. Replacing s by $s - h$ decreases its capped value by between zero and h. It follows that

$$(1-\rho)h \le f_H(S) - F_0(S) \le (1-a)h. \tag{8}$$

Thus $F_0(S) \le f_H(S)$. Both terms in the maximum (5) are at most $(1+\varepsilon)f_H(S)$, and the second term supplies the lower endpoint in (1). This verifies the multiplicative promise globally, including unqueried sets, large sets and the empty set.

The baseline pair (F_0, g_0) is also a valid input of exact curvature κ: its singleton value is ρ and its full-set marginal is a. We may therefore use the algorithm's query and output promises on this baseline. Finally,

$$0 < c \le 1, \qquad c \ge (1-\kappa)\varepsilon. \tag{9}$$

Indeed the denominator defining c is positive, and subtracting its numerator from that denominator gives κ. The second bound follows from $1 - \rho = 2\varepsilon/(1+\varepsilon) \ge \varepsilon$ and $1 - a \le 1$.

The unequal singleton values are essential to this calibration. In contrast, the particular uniformly shifted family

$$U_H(S) = (1-\kappa)|S| + \kappa\big(|S \cap H| + \min\{|S \setminus H|, r\}\big)$$

has optimum k and gives every k-set value at least $(1-\kappa)k$. It cannot witness the smaller fixed-noise factor $\rho(1-\kappa)$. The calibration in Table 1 retains exact curvature while lowering the outside marginal to that product.

2.2 A tail bound valid for all query sizes

Let H be uniform among the k-subsets of V. If $s \le r$, then $f_H(S) = \rho s + (1-\rho)h \le s$ and $g_0(S) = (1-\varepsilon)s$, so $g_H(S) = g_0(S)$ for every H. For $s > r$, the upper bound in (8) implies equality of the oracle answers whenever

$$h \le \theta(s) := \frac{(1-\rho)[(1-\kappa)s + \kappa r]}{1-a} \ge cs. \tag{10}$$

To see this, equality is guaranteed by $(1-\varepsilon)[F_0(S) + (1-a)h] \le (1+\varepsilon)F_0(S)$; solving this inequality gives (10).

Fix an s-set S with $s > r$ and let $j = \lceil cs \rceil$. By (9), $1 \le j \le s$. A revealing answer requires $|S \cap H| > cs$, hence lies in the enlarged event $|S \cap H| \ge j$, even when cs is an integer. If $j > k$ that event is empty. Otherwise a union bound over the j-subsets of S gives

$$\begin{aligned}\mathbb{P}[g_H(S) \ne g_0(S)] &\le \binom{s}{j}\frac{(k)_j}{(n)_j} \le \binom{s}{j}\left(\frac{k}{n}\right)^j \\ &\le \left(\frac{esk}{jn}\right)^j \le \left(\frac{ek}{cn}\right)^j \le \left(\frac{ek}{cn}\right)^{cr}.\end{aligned} \tag{11}$$

Here $(m)_j = m(m-1)\cdots(m-j+1)$ and $(k-i)/(n-i) \le k/n$. The standard estimate $\binom{s}{j} \le (es/j)^j$ follows from $j! \ge (j/e)^j$. The last inequality uses $cn > ek$ and $j \ge cs > cr$. No independence among membership indicators is used. Together with the nonrevealing case $s \le r$, this controls every query size.

3 Adaptive queries and fixed hard instances

Proof of Theorem 1. The construction already verifies the structural promises and optimum. Fix a private seed ω and run the algorithm on g_0. Let $S_1(\omega), \dots, S_m(\omega)$ be its baseline queries, where $m \le Q$, and let $A_0(\omega)$ be its output. All these sets are independent of H. Let B be the event that at least one baseline query has different answers under g_H and g_0. By (11) and a union bound,

$$\mathbb{P}_H(B \mid \omega) \le \eta := \min\left\{1, Q\left(\frac{ek}{cn}\right)^{cr}\right\}. \tag{12}$$

When B does not occur, induction through the successive queries shows that the run on g_H has the same transcript, stopping time and output as the baseline run. The induction uses independence from H only for the baseline queries; it makes no assertion about queries after a divergence. Every feasible output has true value at most k. Pointwise, therefore,

$$f_H(A_\omega^{g_H}) \le f_H(A_0(\omega)) + k\mathbf{1}_B.$$

The baseline output need not have been queried. Nevertheless, with $m_0 = |A_0| \le k$, symmetry gives $\mathbb{E}_H|A_0 \cap H| = m_0 k/n$. Using (4),

$$\begin{aligned}\mathbb{E}_H[f_H(A_0)] &= am_0 + (1-a)\mathbb{E}_H|A_0 \cap H| + b\mathbb{E}_H \min\{|A_0 \setminus H|, r\} \\ &\le ak + (1-a)k^2/n + br.\end{aligned}$$

Average over H and then over ω, and apply (12). This yields (3) on the joint average. Since the collection of H is finite, at least one single H has private-seed expected ratio at most that average. Its oracle is the fixed map (5), independent of the seed, as required. □

4 Matching algorithm and asymptotic regime

Proposition 3 (Standard singleton method). *For any $n \geq k \geq 2$, $0 \leq \varepsilon < 1$, and a valid pair (f, g) with total curvature κ, selecting the k largest $g(\{i\})$ values returns S with*

$$f(S) \geq \rho(1 - \kappa + \kappa/k)\,\mathrm{OPT}(f, k). \tag{13}$$

Proof. Let O be an optimal k-set, extending a smaller optimizer if necessary. Singleton ordering and submodularity imply

$$\begin{aligned}(1+\varepsilon)\sum_{i\in S} f(\{i\}) &\geq \sum_{i\in S} g(\{i\}) \geq \sum_{i\in O} g(\{i\}) \\ &\geq (1-\varepsilon)\sum_{i\in O} f(\{i\}) \geq (1-\varepsilon) f(O).\end{aligned}$$

Enumerate S starting with its largest true singleton. Diminishing marginals and curvature give every later marginal at least $(1-\kappa) f(\{i\})$. Hence

$$f(S) \geq (1-\kappa)\sum_{i\in S} f(\{i\}) + \kappa \max_{i\in S} f(\{i\}) \geq (1 - \kappa + \kappa/k)\sum_{i\in S} f(\{i\}).$$

Combining the inequalities proves the claim. The true singleton order is used only in the analysis; the algorithm queries and sorts noisy values. This is the direct true-objective analysis of the method in Horel–Singer [HS16, Proposition 8]. □

Proof of Corollary 2. Suppress subscripts and set $u = \varepsilon k \to \infty$, $\delta = 1 - \kappa > 0$, and $r = \lceil \sqrt{k/\varepsilon} \rceil$. Then

$$\frac{r}{k} \leq \frac{1}{\sqrt{u}} + \frac{1}{k} \to 0.$$

Eventually $1 \leq r < k$ and $n > k + r$. By (9),

$$cr \geq \delta\sqrt{u} \to \infty, \qquad \frac{cn}{k} \geq \delta u \frac{n}{k^2} \geq \delta u n^{1-2\alpha}. \tag{14}$$

Put $\gamma = (1 - 2\alpha)/2 > 0$. Eventually $ek/(cn) \leq n^{-\gamma} < 1$, so Theorem 1 applies. For the fixed query bound $Q \leq Cn^d$, the disagreement term is at most

$$Cn^{d - \gamma cr} = n^{-\omega(1)}.$$

The remaining error terms satisfy

$$\frac{br}{k} \leq \kappa\left(\frac{1}{\sqrt{u}} + \frac{1}{k}\right) \to 0, \qquad \frac{(1-a)k}{n} \leq \frac{k}{n} \to 0.$$

Applying the finite theorem separately at each sufficiently large n gives (6). Proposition 3 supplies the deterministic lower guarantee. Its gap above $\rho_n(1-\kappa)$ is $\rho_n \kappa / k_n = o(1)$, giving additive matching. When $\inf_n \rho_n > 0$, the central factor is bounded below by a positive constant, so the additive remainders also give relative matching. □

5 Exact finite checks and boundaries

The same transcript proof has a finite alternative that avoids the condition $cn > ek$. Retain $n > k + r$, $1 \leq r < k$ and the other parameters of Theorem 1, and define

$$p = \max_{0 \leq s \leq n} \mathbb{P}_H[g_H(S_s) \neq g_0(S_s)], \tag{15}$$

$$\beta = a + \frac{(1-a)k}{n} + \frac{b}{k}\mathbb{E}[\min\{k - X, r\}], \qquad X \sim \mathrm{Hypergeom}(n, k, k), \tag{16}$$

where S_s is any fixed s-set. Symmetry makes (15) independent of its choice.

Lemma 4 (Exact-tail alternative). *For every Q-query algorithm as in Theorem 1, some fixed H satisfies*

$$\frac{\mathbb{E}[f_H(A)]}{k} \le \beta + \min\{1, Qp\}. \tag{17}$$

Proof. The baseline union bound is at most $\min\{1, Qp\}$. Extend any baseline output of size below k to a k-set and use monotonicity. For every fixed k-set, its expected true value divided by k is exactly (16), by the hypergeometric law of its intersection with H. The same coupling and finite averaging prove (17). □

All probabilities in (15) can be computed by summing

$$\frac{\binom{s}{h}\binom{n-s}{k-h}}{\binom{n}{k}}$$

over feasible h for which the formulas (7) and (5) differ. For $n = 6, k = 3, r = 2$, $\varepsilon = 1/2$ and $\kappa = 1/3$, this gives $p = 1/20$ and $\beta = 359/540$. Exhaustive optimization of deterministic adaptive decision trees of depth zero, one and two gives respective best uniform-prior expected ratios $359/540$, $7/10$ and $13/18$, each satisfying (17). A randomized policy is a mixture of deterministic policies, so it cannot improve those prior optima. The portable supplementary script reproduces these checks and verifies additional all-subset and count-based instances. These finite observations supplement the analytic proofs.

The curvature-one endpoint, the transition with bounded εk, and dense cardinality remain open directions for a fuller joint-parameter characterization. The case $k = n$ provides a boundary condition: returning V is optimal. Further questions include exact finite minimax factors and high-probability guarantees on fixed adversarial inputs.

Computational and AI assistance. Codex assisted with the construction, proof development, exact finite checks, literature comparison and drafting. A separate model-context pass checked the mathematical arguments.

References

[BCF+25] Kshipra Bhawalkar, Yang Cai, Zhe Feng, Christopher Liaw, and Tao Lin. A unified approach to submodular maximization under noise, 2025. arXiv:2510.21128v2.

[HS16] Thibaut Horel and Yaron Singer. Maximization of approximately submodular functions. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[HS17] Avinatan Hassidim and Yaron Singer. Submodular optimization under noise. In *Proceedings of the 2017 Conference on Learning Theory*, volume 65 of *Proceedings of Machine Learning Research*, pages 1069–1122. PMLR, 2017.

[SVW14] Maxim Sviridenko, Jan Vondrák, and Justin Ward. Optimal approximation for submodular and supermodular optimization with bounded curvature, 2014. arXiv:1311.4728v3.